

Electric Vehicles :

23 July 2015
Venus

Introduction : An electric vehicle is a vehicle that uses one or more electric motors for propulsion instead of an internal combustion engine (ICE).

Basic Principle :

Electrical Energy $\rightarrow$ Electric Motor $\rightarrow$ Mechanical Energy

Wheels Motion $\rightarrow$ Rotational Energy

ICE $\rightarrow$ Chemical Energy $\rightarrow$ Heat $\rightarrow$ Mechanical $\rightarrow$ Rotational motion

Basic Component of EV :

1) Battery - Lead Acid / Lithium Ion

1900-1920 - EV very popular (Golden Era) after 1920 - fuel storage and road were built - shifted to ICE - EVs got drawn back.

Function - To store electrical energy - supply DC power to motor - determine range.

Different EVs - figure out battery pack size of EVs

Tata Motors - CUSKRV

2) Electric Motor - IM, BLDC, PMSM

Power loss - 3% of the delivered energy lost as heat in conductors and transformers

- Causes: 1) High charging current
2) Long cable length (higher impedance drop in cables)
3) Overloaded conductors
4) Harmonic currents

Effect: 1) Increased electricity cost
2) Heating of cables and transformers
3) Reduced overall efficiency

- Solution: 1) Reduced feeder length where possible
2) Improve power factor
3) Use large conductors
4) Install charging station closer to substation

Overloading of transformers: It occurs when the load increases the TL (cable capacity)

Effects: - Faster insulation aging
- Overheating
- Reduced life of TL
- Increased maintenance

- Solutions: 1) Using smart charging
2) Load management
3) Upgrade TL
4) Install more TL (distribution TL)

The Power Quality degradation:

It refers to maintaining a clean stable and available supply of electrical power with correct V , f and waveform

- Causes: Harmonics, voltage sag, fluctuations, upset EV charging units, poor power factor, unbalanced loading

Effects: Reduced equipment life, equipment overheating, sensitive electronic malfunction

Solution: Smart grid technology, Energy storage system, harmonic filters, demand response program, demand response can improve power monitoring

Basic Vehicle Performance:

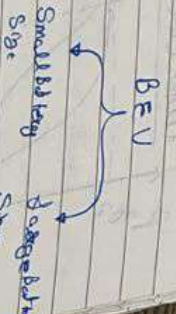
- Parameters: 1) Engine power
2) Range
3) Efficiency
4) Acceleration
5) Power of motor, slope, grade etc

2) Range - Capacity of fuel

- Acceleration
- Load (weight)
- Road Condition
- Driving style (speed etc)

→ driving cycle - Urban drive cycle

- 3) Efficiency - How efficient the car is.
4) Acceleration - driving up or down a slope.
5) Power of motor, slope, grade etc



Impact of modern drive train of power supplies

Drive train - Collection of component during stroke
 Regen - EVHEV
 Impact - live or -ve
 Power supply - 400V.

1700W EV charging simultaneous - 100W - 170W EV.

Voltage Instability - If the condition is such that the supply voltage changes from its nominal value due to sudden or excessive loading.

5% change of deviation - for 230V $\rightarrow$ 399-441V.

Cause of voltage stability -

1.) Sudden Charging / load increase - It increases load and causes feeder voltage to drop.

2.) Long distribution network - dead near to the end will have low impedance and farther away will have high impedance.

Consumer located far from distribution will receive large voltage drop.

3.) Load distribution - Old and underused distribution lines are underused conductors and overloaded feeders can't maintain voltage during EV charging demand.

Fast DC charger - Rapid charging in charging power stations can produce sudden voltage fluctuations.

Effects - Reduced Motor efficiency

2) Malfunction in electronic equipment

3) Slower charging speed

Fixes / Improvements - 1. Smart Chargers

2. Upgraded infrastructure
 3. Smart voltage regulators

Harmonics - Causes - EV chargers contain power electronic converters

Effects - If overloading, cable overheating, additional power losses and reduced motor efficiency

Solution - UPF function of active power filters.
 High power factor chargers.
 Voltage sag

Voltage Sag - Causes - fast charging station switching on - creates immediate drop.

Sudden increase in EV charging load

Effects - Battery charging errors after
 2) Charger interruptions.
 In Solutions -

Solutions - 1. Soft charging starter
 2. Dedicated feeders
 3) Energy storage system.
 4) Dynamic voltage restorers.

Dynamics Model for Vehicle:

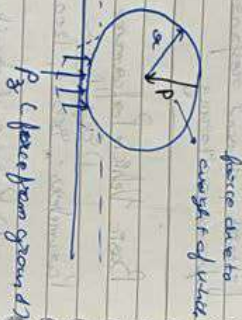
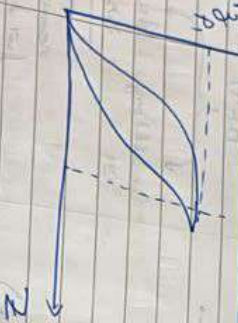
Using the second law of motion $\rightarrow \frac{dV}{dt} = \frac{F}{m}$

friction $\rightarrow$ μ = coefficient of friction
 μ = rolling friction
 μ = opposition due to gradient (dry)

To move the vehicle up the slope we have to produce force to exceed f_r & f_g

1.) Rolling Resistance:

When Car is at standstill:



There can't be the same unloading loading conditions for the tyre of a car. There will be hysteresis that will cause different dependence of tyre causing different path.

When Tyre Perforating:

Tyres gets more deformed from the speed to maintain a bit more friction to make the car stable and grip the



Factors affecting rolling resistance:

- 1) Material of tyre - hard or soft.
- 2) Tyre structure - T_g
- 3) Tyre Temperature
- 4) Tyre Inflation pressure - To reduce rolling resistance - higher pressure
- 5) Road roughness - Mechanical road - presence of liquid on road

Acceleration of a Vehicle

Using Newton's 2nd law of motion vehicles motion can be expressed as:

$$\frac{dV}{dt} = \frac{\sum F_e - \sum F_r}{M_s}$$

V = vehicle speed

$\sum F_e$ = total tractive force

$\sum F_r$ = total resistance force

M_s = mass of the vehicle

G = mass factor of the vehicle when converting motion to translational motion.

The moment force produced by the lateral shift of resultant ground force is called as rolling resistance factor.

P = normal force acting on the centre of the rolling wheel (Unit Newton)

a = Deformation of tyre (Unit m)

M = Mass of vehicle (kg)

g = Acc. due to gravity

$$T_{\text{or}} = P a = M g a$$



To keep the wheel rolling (for) force acting on the centre of the wheel is required to balance this rolling resistance moment.

$$F_{\text{or}} = \frac{T_{\text{or}}}{a_{\text{dyn}}} = \frac{P a}{a_{\text{dyn}}} = P \frac{a}{a_{\text{dyn}}}$$

dynamic radius of tyre (metres (m)) rolling resistance coeff.

Rolling Resistance $\rightarrow F_{\text{or}} = \frac{T_{\text{or}}}{a_{\text{dyn}}} = \frac{M g a}{a_{\text{dyn}}} = M g f_{\text{or}}$

$$T_{\text{or}} = P a = M g a$$

2) Aerodynamic Drag:-

→ Skin effect - Vehicle for air flow as curved motion instead of flat one suffering so less resistance to air vehicle

Shaped vehicle

If front area more - more drag experienced by the vehicle more curved



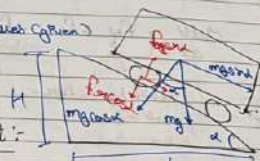
$$\text{Skin effect: Aerodynamic Drag} = \frac{1}{2} \rho C_d A_f v^2 = F_d$$

ρ = Density (kg/m³)

C_d = drag coefficient } constant values (given)

A_f = vehicle front area

v = velocity



3) Opposition due to gradient:-

Opposition or gradient resistance can be expressed as:-

$$F_g = M g \sin \alpha$$

4) Acceleration Resistance:- (F_a)

$$F_a = \left(M + \sum \frac{I_{\text{rot}}}{a_{\text{dyn}}} \right) \frac{dv}{dt}$$

I = inertia of moving component

kg m²

M = mass of vehicle

For going uphill we also consider normal force forces
 We also consider $\alpha = H/L$ for small uphill gradient

$$f_{\text{net}} = f_{\text{normal}} = f_{\text{net}} + f_N + f_g + f_a$$

using second law

$$\frac{dv}{dt} = \frac{f_{\text{net}} - \sum f_{\text{resistance}}}{M}$$

$$M \frac{dv}{dt} = f_{\text{t}} + f_{\text{tr}} - f_{\text{na}} - f_{\text{rg}} - f_a - f_{\text{tr}} - f_g$$

tractive force both front wheels
 tractive force both front wheels
 tractive force both front wheels

tractive force both front wheels

tractive force both front wheels

$$W_L = M g L \cos \alpha - (T_{\text{ax}} + T_{\text{ay}} + f_{\text{tr}} h_w)$$

$$+ M g h \sin \alpha + M g \frac{dy}{dt}$$

drag force

drag force

drag force



By summing the moment of all the forces by point R the normal force in the front axle determined by:

$$W_f = M g L \cos \alpha - (T_{\text{ax}} + T_{\text{ay}} + f_{\text{tr}} h_w + M g h \sin \alpha + M g \frac{dy}{dt})$$

$$M g L \cos \alpha - (T_{\text{ax}} + T_{\text{ay}} + f_{\text{tr}} h_w + M g h \sin \alpha + M g \frac{dy}{dt})$$

L

For passenger car $h_w = h_g$ and take $\cos \alpha \approx 1$

$$= M g L \cos \alpha - h_g [f_{\text{tr}} + f_g + M g \frac{dy}{dt} \cos \alpha + M g \frac{dy}{dt}]$$

$$W_f = M g L \cos \alpha - h_g [f_{\text{tr}} + f_g (1 - \frac{\sin \alpha}{h_g})]$$

$$f_{\text{tr}} = W_f \times \mu$$

Slip factor: The slip bias tyre surface, if muddy

$$S_a = \frac{C_{\text{DR}} \sigma_{\text{tragn}} - V}{C_{\text{DR}} \sigma_{\text{tragn}}}$$

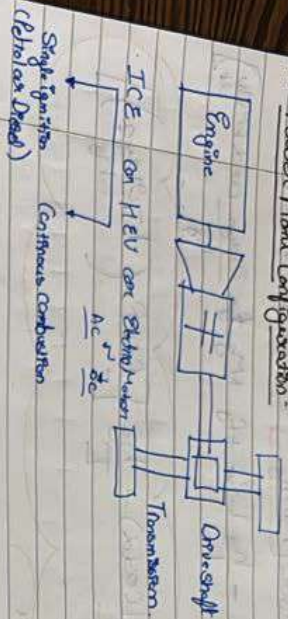
car tragn.

When the tractive effort of wheel exceeds, the maximum tractive effort limit imposed by adhesive capability of the tyre and the ground.

The driven wheel will spin on the ground.

Power - Torque Characteristics

Power Plant Configuration

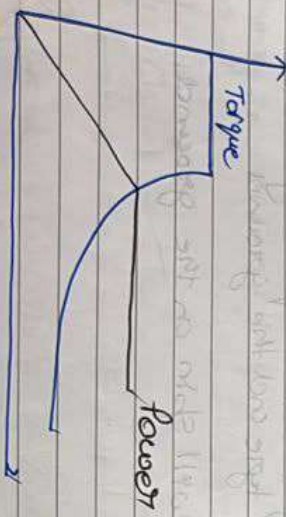


Petrol
(advantage)

Diesel
(advantage)

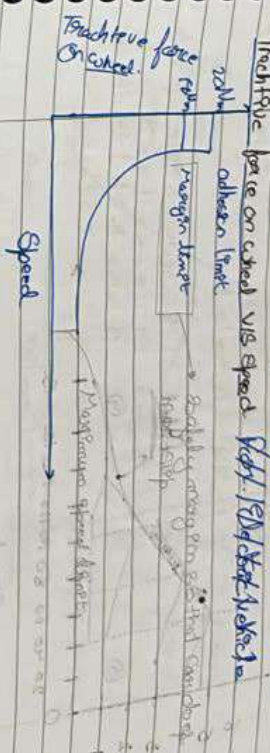
- High energy density
- Low consumption time
- Better performance
- discharge
- Higher fuel consumption
- Quality of fuel required high

Torque characteristics of electric motor



11/10/20

used app. Method performance characteristics of power plant
Power plant power output over full speed range
Consequently the torque varies hyperbolically w.r.t. speed.



Maximum torque is limited by -
1) Adhesion force - (Pne. rolling, grading, drag - cannot attain traction limit)
2) Traction force - (Pne. rolling, grading, drag - cannot attain traction limit)
3) Traction force - (Pne. rolling, grading, drag - cannot attain traction limit)

Velocity is limited by -
1) Valve open and close limit for ICE
2) Fuel burn time
3) Torque limit (Piston are down)

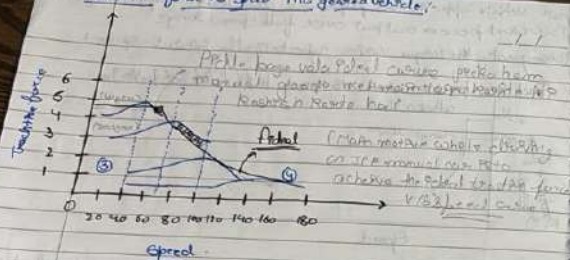
For Motor velocity is limited by
1) Motor heating limit
2) Insulation breakdown due to overheating (breakdown)
3) High centrifugal force at high rpm.

Common features:-
1) Mechanical transmission system
2) Vibration limits
3) Bearing limits.

Common features:-

- 1) Mechanical transmission system
- 2) Vibration limits
- 3) Bearing limits.

Traction force vs speed for geared vehicle



Various Indices (Performance Indices)

1) Torque increase and Power: Various indices are used to characterize different types of engines.

1) Torque increase indices =

$$T = \frac{T_{max}}{T_n} \rightarrow \text{Torque at which power is max.}$$

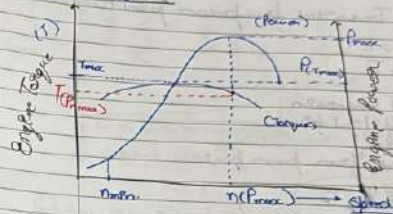
$T_n = T(P_{max})$

2) Engine speed indices

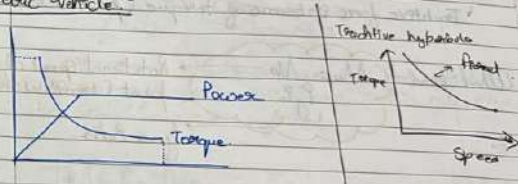
$$V = \frac{V_n}{V(T_{max})} \rightarrow \text{Speed at max torque}$$

The highest value of the product $T \times V$ - better engine power at low and medium speed.

This arrangement allows less frequent gear changing for IC Engine.



For Electric Vehicle



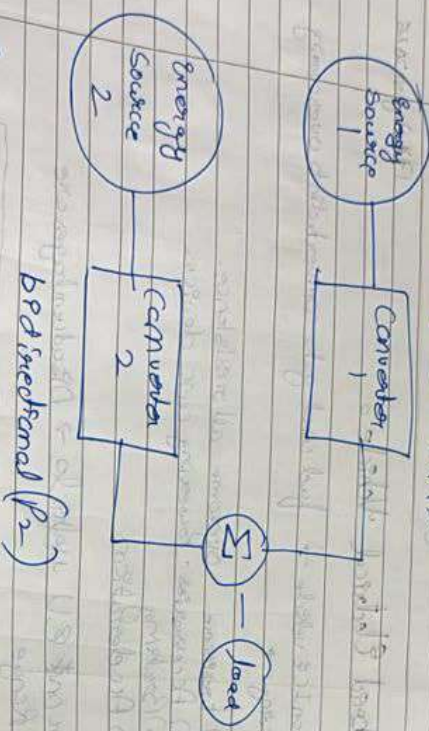
With the maximum available engine power P_{max} and speed V_n , ideal torque vs speed.

$$T_{ideal} = \frac{P_{max}}{V} \quad , \quad T_{effective} = \frac{P_{max}}{V} \times \eta_{total}$$

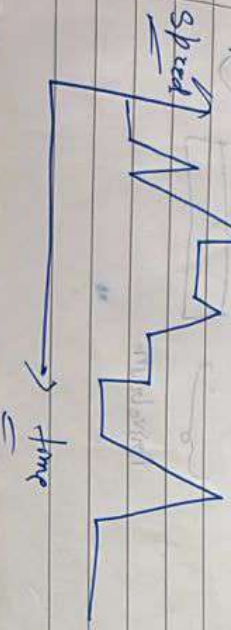
Efficiency of total drivetrain

HEV $\rightarrow$ ICE + Battery
 Fuel cell + battery
 all load power + battery
 P (fuel cell) + battery
 battery + battery

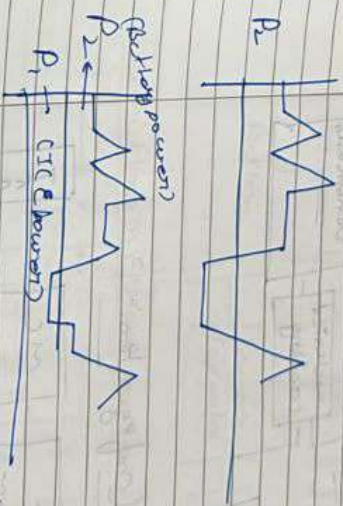
Power transfer :- for HEV
 Controlled Power (P)



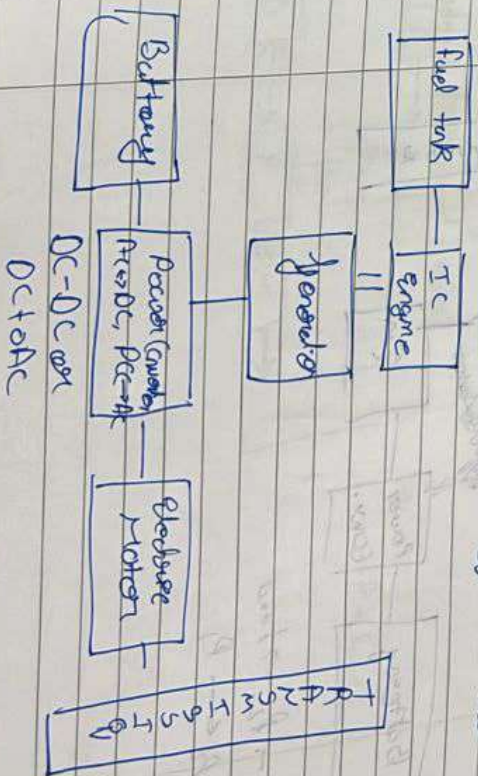
(1) $P_1 \rightarrow$ load
 $P_2 \rightarrow$ load
 $P_1 + P_2 \rightarrow$ load
 In series $P_1 \rightarrow P_2 \rightarrow$ load
 Powering cycle :- (How your speed changes with time when power is applied)



(2) $P_2 \rightarrow P_1 + \text{load}$
 Battery Engine
 $P_1 \rightarrow P_2 + \text{load}$
 Engine battery



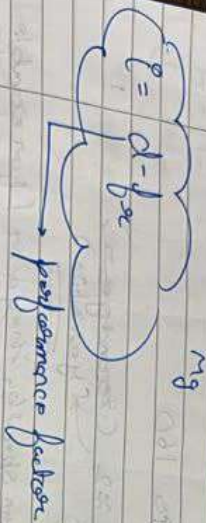
Configuration :- Series Configuration - long range
 hence m bigger value



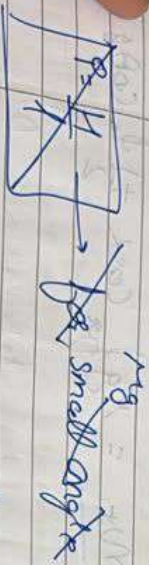
2.1) Acceleration $\dot{a}$:-

$$\frac{T_p}{I_p} \frac{g}{g_0} \eta + \frac{M_g}{I_p} \cos \alpha + \frac{1}{2} \rho_a \cos \alpha V^2 + M_g \dot{a}$$

$$P = T_p g_0 \eta + \frac{1}{2} \rho_a \cos \alpha V^2 + M_g \dot{a}$$



$$d = T_p g_0 \eta + \frac{1}{2} \rho_a \cos \alpha V^2$$



$$S_{md} = \frac{d - f_x^2}{1 + f_x^2} \left(1 - d^2 + f_x^2 \right) \left\{ \begin{array}{l} \text{for large } d \\ \text{for small } d \end{array} \right.$$

03/05/2008, 10:45 AM, 11/11/2008

3) Acceleration Performance :-

$$a = \frac{f_t - f_w - f_b}{M_g} = \frac{g}{S} (d - f_x)$$

$$S = \text{statistical factor}$$

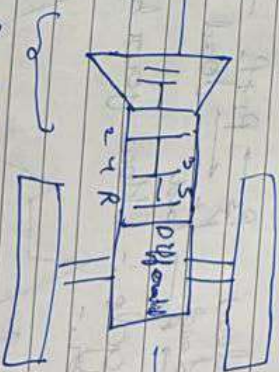
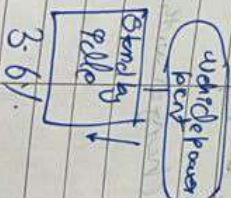
$$S = 1 + \frac{T_w}{T_{w, \text{stagn}}} + \frac{\rho_a g_0^2 I_p}{M_g}$$

Hybrid Electrical Vehicle :-

For on ICE vehicle $\rightarrow$ fuel used gets converted into overcoming following $\rightarrow$

- 1) Friction - overcome all resistance.
- 2) Acceleration - overcoming air resistance.
- 3) Braking
- 4) Acceleration

For on EV vehicle $\rightarrow$ Disadvantages are



Efficiency $\rightarrow 70-90\%$
 Braking $\rightarrow 20-30\%$

Mathematical Modelling:-

1.) Torque - Transmitted from power plant to driven wheel

$$T_w = I_g I_a \eta_t T_p$$

I_g = gear ratio of transmission
 I_a = gear ratio of final drive
 η_t = total efficiency
 T_p = torque output from power plant

$$F_t = \frac{T_w}{r_{\text{wheel}}} = \frac{I_g I_a \eta_t T_p}{r_{\text{wheel}}}$$

Traction force in terms of torque (friction resistance)

Wheel Speed - $N_w = \frac{N_p}{I_g I_a}$ • Rotational speed of power plant (from rotational)
 • Gear Ratio

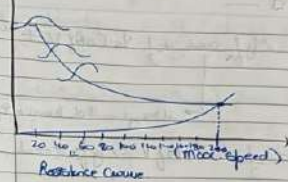
Translational Speed of wheel -

$$V = \frac{\pi N_w r_{\text{wheel}}}{60} \text{ m/s}$$

Vehicle performance is affected by

- 1) Maximum Speed (Cruising Speed)
- 2) Accelerability
- 3) Acceleration

Torque



Delia say max speed is 160

At 60 Speed - $\frac{160}{60} \approx 2.67$ (2nd gear ratio)

Maximum speed by the gear (for example this was second gear so gear ratio must be of 2 to attain max speed of 160.

$$\frac{T_p I_g I_a \eta_t}{r_{\text{wheel}}} = \frac{Mg}{\cos \theta} + \frac{1}{2} \rho A C_D V^2$$

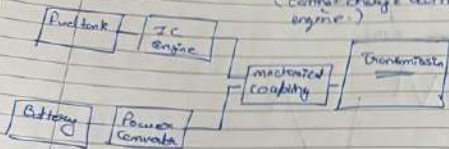
Rolling Resistance Drag

$$V_{\text{max}} = \frac{\pi N_{\text{max}} r_{\text{wheel}}}{60}$$

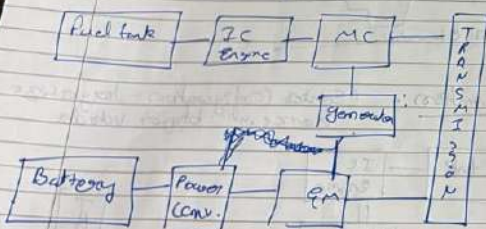
Maximum speed in km/h when gear ratio is

Gen → charges bat heavy and continues to work

Parallel Configuration:- Used P.M. primary vehicle (cannot charge but transform engine)

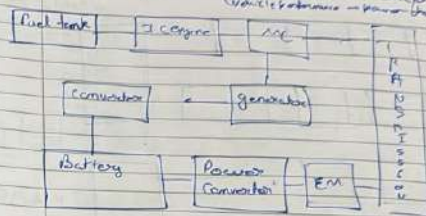


Series-Parallel Configuration:-

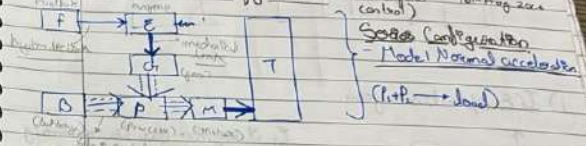


$P_1 \rightarrow P_2 \rightarrow \text{load}$
 $P_2 \leftarrow P_1$

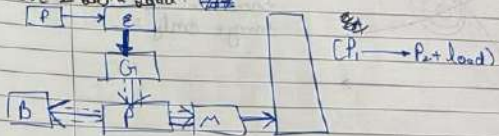
Complex design:- $\begin{cases} \text{Social} \\ \text{Eco} \\ \text{Vehicle performance} \end{cases}$ → $\begin{cases} \text{Inputs, constraints} \\ \text{Numerical} \\ \text{power distribution} \end{cases}$

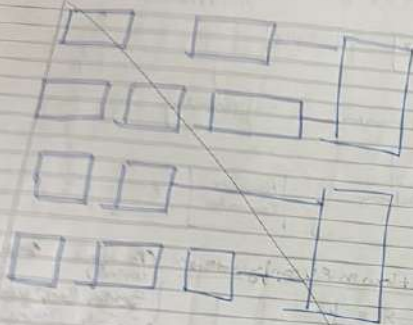


Power Flow in EV configuration:- (Power flow control) 18th Aug 2021



Mode 2 light load





1) ICE operation region.

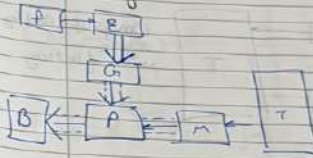


Eg. If engine can't be running at 200 km/h the ICE should be used at a certain range only i.e.

2) ICE operating line

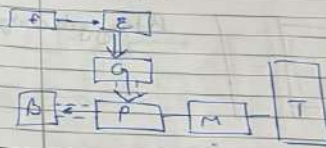
3) battery operating - Not charge more than a certain level and not discharge more than a certain level.

Mode 3 Braking:-



(Load $\rightarrow$ $P_{in} \rightarrow P_{out}$)
Regenerative Charging

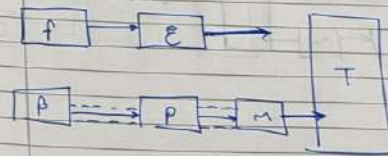
Mode 4 Standstill:-



Engine charges battery at standstill

Parallel Configuration:-

Mode-1 Start up / full throttle / acceleration



Use both battery and engine to supply load.

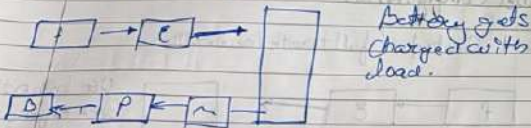
Mode-2 (Normal Acceleration)



Mode-3 Braking



Mode-4 Light decelerating



Series Parallel

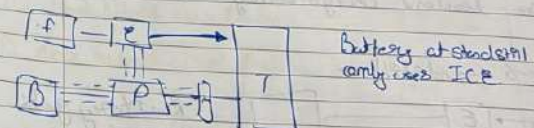
Mode-1 Start-up



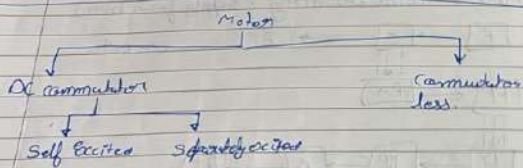
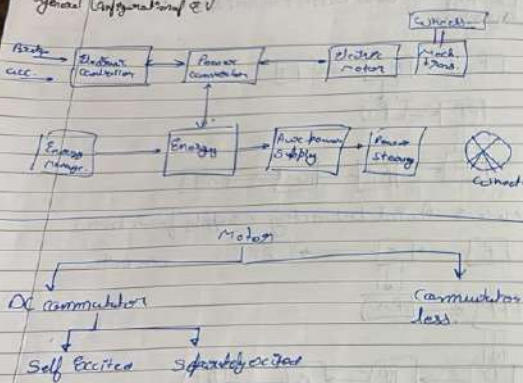
Mode-2 Full acceleration



Mode-3 Normal acceleration



General Configuration of EV



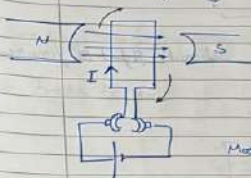
DC Motor Drive :-

- High speed operation
- Adjustable speed
- Regenerative braking
- Braking & coasting

Problems to low specific power motor
(more weight, low power)

25. Reg. drive

(FLH & reg. drive)
(PR & reg. drive)



Torque produce :-

$$T = BIL \cos \theta$$

More coil side, more torque.

$$V_a = I_a R_a + K_e \phi \omega_m$$

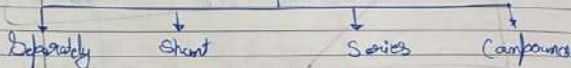
$$E = K_e \phi \omega_m$$

$$T = K_t \phi I_a$$

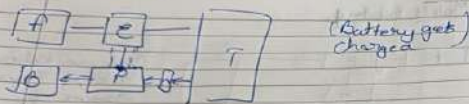


If permanent magnet is not used and wound electromagnet is used.

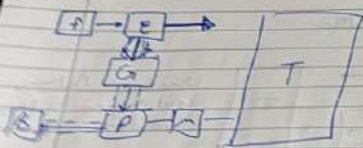
General Motor



Mode-4 Braking

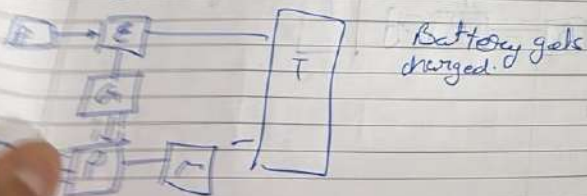


Mode-5 Battery charging in normal conditions

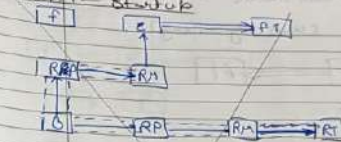


If power demand is more then use ICE to both and when it becomes less use ICE to both and charge battery

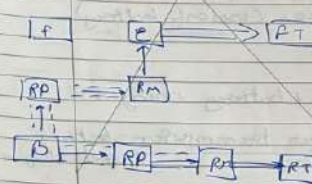
Mode-6 Standstill :-



Complex System :- (Front and Rear Transmission)



Mode-7 - Full Acceleration (uses only battery)

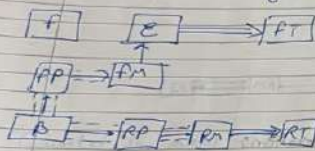


These are different modes to operate complex hybrid system

- ① front hybrid Rear Electric $\rightarrow$ ICE heavy
- ② Rear hybrid - front Electric $\rightarrow$ battery heavy

→ Front hybrid power electronic

Mode-1 - Start up (uses only battery)

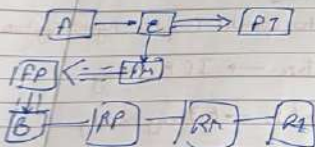


Mode-2 - Maximal Acceleration (uses only battery)

(Same as start-up)

Mode-3 - Vehicle propels + battery charges

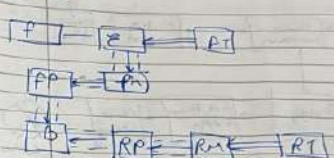
No use of ~~maximal transmission~~ battery gets charged



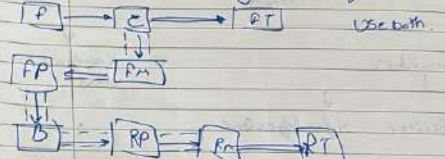
Mode-4

Full Acceleration - use only battery, No use of engine

Braking - Uses both transmissions to charge battery



Mode-5 - Axial Balancing (supply from both)



Rear Hybrid Power Electronic

In Mode-1 - Start up (Battery to charge transmission)

Mode-2 - ~~both~~ fuel give power to both FT, RT and battery also provides to FT.

Mode-3 - Vehicle propels + Bat. charging

f -> FT
f -> Battery

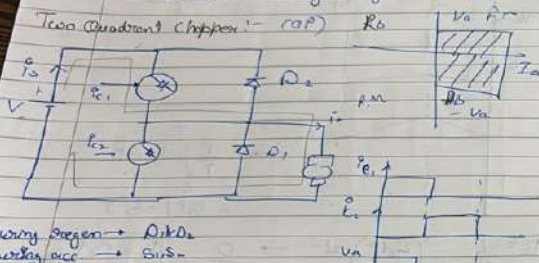
Mode-4 - Light load -> f -> FT

Mode-5 - Braking -> FT -> B
RT -> B

Mode-6 - Axial Balancing f -> FT, f -> B

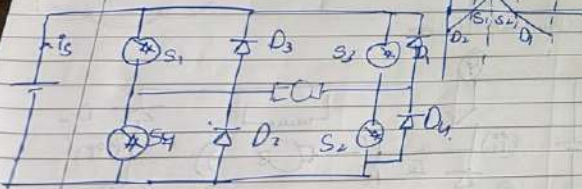
Current will flow from (D, P, T, G) when switch is closed
 When switch is open → Current direction through armature
 remains same but voltage changes → power

Two Quadrant Chopper:- (RPF)



Driving signal → D_1 and D_2
 Driving sig → S_1 and S_2

Class E Four Quadrant



Induction Motor:-

→ Small size
 → Low cost
 → High efficiency
 → High efficiency (control) → Direct Torque Control
 → Field Oriented Control → This is used more often due to lack of torque
 → Cost effective → controlled using VVVF C Variable Voltage - Variable frequency control

FOC - Field Oriented Control (Vector Control)

Complexity due to non-linearity of Induction motor, VVVF method is not used and FOC is more predominant

Principle of Induction Motor:-

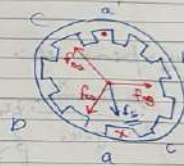
$$I_{as} = I_m \sin(\omega t) + I_{as} = I_m \sin(\omega t)$$

$$I_{bs} = I_m \sin(\omega t + 120^\circ)$$

$$I_{cs} = I_m \sin(\omega t - 120^\circ)$$

$$I_s = \frac{3}{2} I_m e^{j(\omega t - \theta)}$$

(complex)

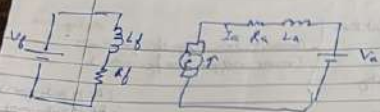


Resulting Magnetic field:-

when $\omega t = 0^\circ$ $I_{as} = 0$, $I_{bs} = -I_m/2$, $I_{cs} = I_m/2$
 at $\omega t = 90^\circ$ $I_{as} = I_m$, $I_{bs} = -I_m/2$, $I_{cs} = -I_m/2$

$$I_s = 0$$

$$\cos \phi = \frac{P}{Q}$$



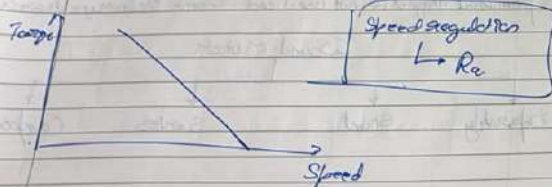
In separately excited DC motor separate supply is provided to field and armature.

V_f constant, ϕ = constant

$$T = K_e \phi \left[V_a - K_e \phi \omega_m \right]$$

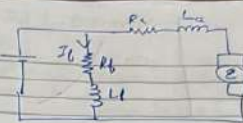
$$T = \frac{K_e \phi V_a}{R_a} - \frac{(K_e \phi)^2 \omega_m}{R_a}$$

It's a linear equation with only changing factor ω_m & T & ω_m



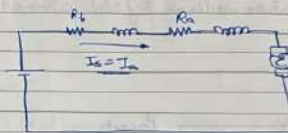
Application - We require sep excited DC motor for good speed regulation over adjustable speed.

Shunt:-



Everything same as separately excited

Series Excited:-



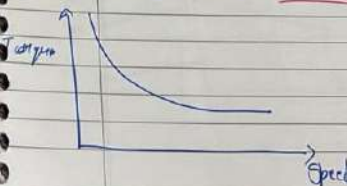
$$\phi = f(I_a)$$

$\uparrow T = K_e \phi I_a \uparrow$ (This provides high torque for heavy load)

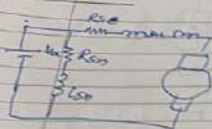
$$T = \frac{K_f K_e V_a^2}{(R_a + K_f K_e)^2}$$

$$T = \frac{K_f K_e V_a^2}{(R_a + K_f K_e \omega_m)^2} \quad (\text{parabolic})$$

Its disadvantage is when load is less speed is very high.

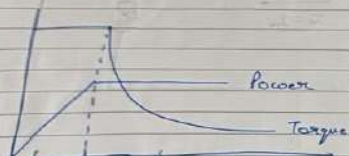


Compound:-



(See lag and short shunt)

Constant armature voltage & field current control:



Armature control
 $V_a = I_a R_a + E$
 $E = K \phi \omega_m$
 $T = K \phi I_a$

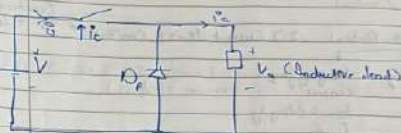
field control
 For constant torque ω_m is constant
 armature voltage control ω_m is constant
 PE devices (choppers)

For field control we get a constant power output

Chopper Control for DC motor:-

Advantages are:-
 - High efficiency
 - Light weight
 - Quick response

Disadvantages:-
 - Electromagnetic Interference
 - Chopper $\rightarrow$ high frequency operation $\rightarrow$ high switching losses
 - Harmonic distortion

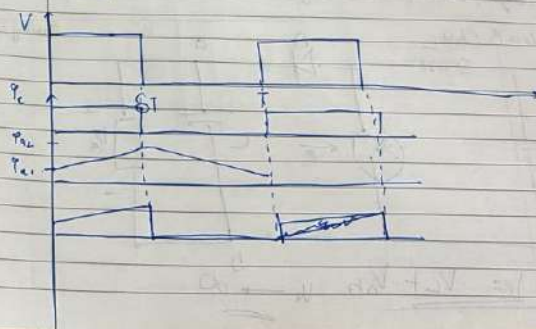


Control Signal:-

Transistor - Base Current
 GTO - Gate Current
 Thyristor - Gate Current

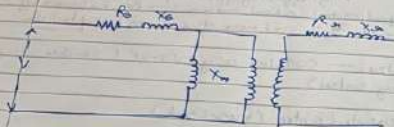
MOSFET - Gate Source Voltage

$B = \text{duty}$



Steady State Performance

Eq. (1) - Subst. L & R



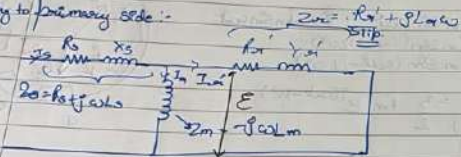
Equation:

$$a_{11} = \frac{R_a}{s}$$

For squirrel cage motor

$$X_m = 0$$

Referring to primary side:



$$I_a = \frac{V}{Z}$$

$$I_a' = \frac{Z_m}{Z_m + Z_{sc}} I_a$$

$$R_a' = a_{11} R_a$$

$$X_m' = a_{11} X_m$$

$$P_{\text{output}} = 3 I_a'^2 \frac{R_a'}{s}$$

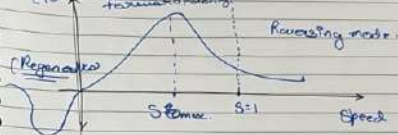
$$T = \frac{P_{\text{mech}}}{\omega}$$

Revolving magnetic field rotates at synchronous speed

$\omega_{ms} = \text{Synchronous speed}$
 $\omega_{ra} = \text{Rotor speed}$
 $\omega_{sl} = \text{Slip speed}$

$$\omega_{ms} = \omega_{ra} - \omega_{sl}$$

Torque Slip Speed Characteristic



$$\omega_{sl} = \frac{2}{p} \omega (1-s)$$

When $s=1$, $\omega_{sl}=0$

At $s > 1$, $\omega_{ms} = -ve$ we enter reversing mode.

When motor speed > Synchronous speed $\rightarrow$ Regenerative mode.

For load current to flow, carrier levels E_{BOP} & $E_{\text{BOP}} + E_{\text{BOP}}$ impose current conduction by providing constant voltage from external sources.

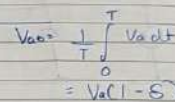
1) Time Rate Control, 2) Current Limit Control $\Rightarrow \delta \rightarrow I_L$

For a factor
 Tan $\frac{1}{2}$... $\frac{1}{2}$... $\frac{1}{2}$...
 are given ...
 O/p voltage

This was first quadrant chopper control having +ve speed, torque, current and voltage

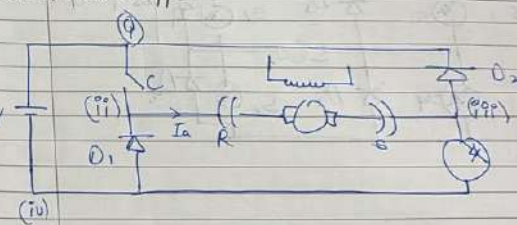


$$= V_L + V_{ab} \quad V_b \rightarrow \infty$$



Advantages - (i) Ripples are less as are voltage. $V_a = \frac{V}{(1-\delta)}$

Two Quadrant Chopper



$$\begin{array}{r} 2.9546 \\ \times 26 \\ \hline \end{array}$$

(Davenport)

Mid-20
End-40

End - 40
Ass - 3

Drugs - 10

File-5

Feb-20
5-5-15

gals viva beforu

0	0.1
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24th July 25

→ moved to EC

3

Integrated circuit)

parents.

points

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Abstract TC7A

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Indice Final

L



What is microprocessor - It is a CPU on a chip which contains ALU, CU and Registers — difference from

Definition: It is a semiconductor component designed by using VLSI design technology and it contains ALU, CU and registers of a CPU in a single package.

* Microprocessors are actually just CPU on a chip.

Note # - Micro processor memory is connected externally

* (ii) Registers inside the processor are used to hold the data temporarily therefore it does not constitute as memory

→ Computer only understands zero and one. - Bit is a binary digit (0 or 1) - Nibble is 4 bit - Byte is 8 bits.

↳ Viva question

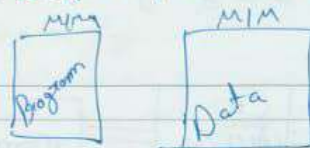
Word length - Depends on type of processor - No. of bits that can be processed parallelly at a time

History of microprocessor:

1971 → Intel 4004 → 4 bit
1972 → Intel 8008 → 8 bit
1974 → Intel 8080 → 8 bit
1974 → " 8085 → 8 bit - Gate, ICS, PSU
1978 → Intel 8086 → 16 bit - ICS
" 8088 → 16 bit
80186 — 80386 → 32 bits.

Pentium } → 64 bits.
95, 95, 97, 99 }

ii) Harvard Architecture - Different Memory Space.

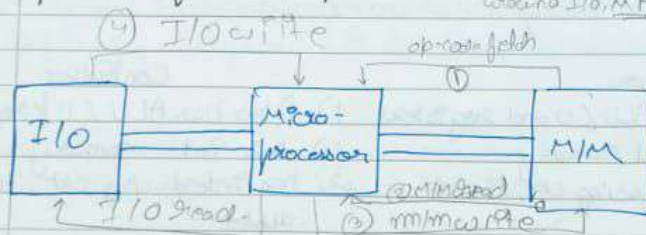


8085 and 8086 - Use Von-Neumann architecture.

8081 - Harvard Architecture.

Very Imp.

Basic Operation of a Microprocessor :- Entire system revolves around I/O, M/P, M/M.



Micro processor can communicate with memory and I/O devices

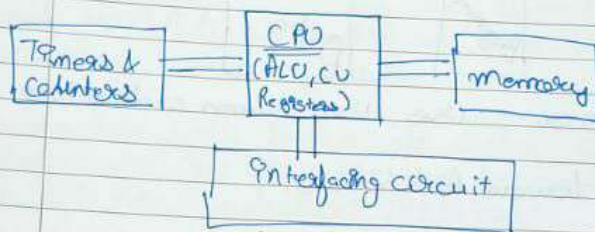
Use assembler to translate from assembly language to machine level by
 MOV: A ← B or MOV: A ← D (data).

Data can program some fetch marker - Use how to having the
 instruction (program) from memory to processor
 called as op-code fetch from memory to processor.

After decoding of op-code fetch, we can either send data or send
 data from memory or send data or receive data
 from I/O devices. Hence that write - memory read,
 memory write - I/O read I/O write.

So here we see following Von-Neumann architecture in
 which program and data both are available in same memory
 space.

microcontroller



Processus

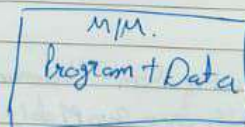
- Controllore

81st July 26

- diff. architecture

fabrika paka:-

- (i) Von Neumann or Princeton Architecture - Same memory



Program is set of instructions that is present in memory and microprocessor only understands binary lang. So, we have to design a language called assembly language. Example MOV A, B where A & B are two registers, which means move content of B in A.

The Assembly will form a code called op-code micro-processor requires instructions to perform a task, it will not take any decision on its own.

So first operation of processor is to bring the instruction from memory into it. After bringing the instruction into the processor, it will decode it and perform various tasks.

Definitions:-

- 1) Op Code Fetch :- Reading / Accessing Instruction (Operation - Code) from memory into processor.
- 2) Memory Read :- Reading / Accessing data from memory into processor.
- 3) Memory Write :- Sending / Transferring data from processor to memory.
- 4) I/O Read :- Reading / Accessing data from port or device I/O processor.
- ★ bus - indicates connection of I/O devices.
- 5) I/O Write :- Send / Transferring data from processor to O/I port device.

In ① & ② both are read operation but in ① we are bringing instruction from mem in ② we are bringing data.

Current across relay / operating coil = $(I_1 - I_2)$
 Current across restraining coil = $(I_1 + I_2)$
 And across operating coil = $\frac{1}{2} (I_1 + I_2)$
 And $\therefore$ operating coil = $\text{Max} (I_1 - I_2)$

At balance condition
 $T_0 = T_R$ (Balance operating coil)

$$\text{Net } (I_1 - I_2) = \text{Net } (I_1 + I_2)$$

$$\frac{2(I_1 - I_2)}{(I_1 + I_2)} = \text{Net}$$

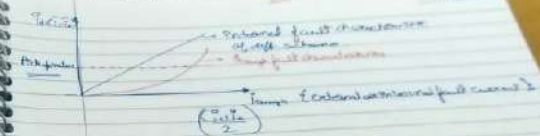
No
 ↓
 D.R.S.

*On the basis of loc. ratio
 we can find the operating condition*

Through fault - One end that passes through the protection zone

$$I_{\text{through}} = \frac{I_1 + I_2}{2}$$

Characteristics of differential scheme



Principle is based on if pick value then after time & hence this fault can take time pickup value

• works for internal fault

In case of internal fault, there are losses hence operation is within spec. If external then no element causes operating time to be more or not even register

